Business Process Modeled

The business process modeled is the point-of-sale (POS) transactions at the P&C grocery store chain. This includes capturing sales data from customer purchases at the checkout, such as item details, quantities, prices, and transaction metadata, to support basic sales analysis like total sales by product, store, date, or customer. It can help to address, for instance, what are total sales by product category per day? etc.

Grain of the Fact Table

The fact table is at the transaction line-item level (each row represents one product sold in a specific transaction). This is the most granular level for POS data, allowing aggregation up to transaction, day, store, or product levels. It's additive for measures like quantity and price, supporting detailed analysis without losing item-specific details.

Assumptions

1. The receipt includes standard attributes like transaction ID, date and time, store details, item code, quantity, price, tax, etc.. Additional attributes such as product, category, brand, are assumed based on typical grocery point-of-sale (POS) systems and dimensional modeling.
2. The receipt also captures customer information which supports a customer dimension.
3. And, in general, promotions, discounts, or coupons should be considered at the item level to enable analysis of promotional impacts on sales.
4. Product hierarchies and store hierarchies in the source system should also be considered for richer analysis.

Questions for the P&C Executives

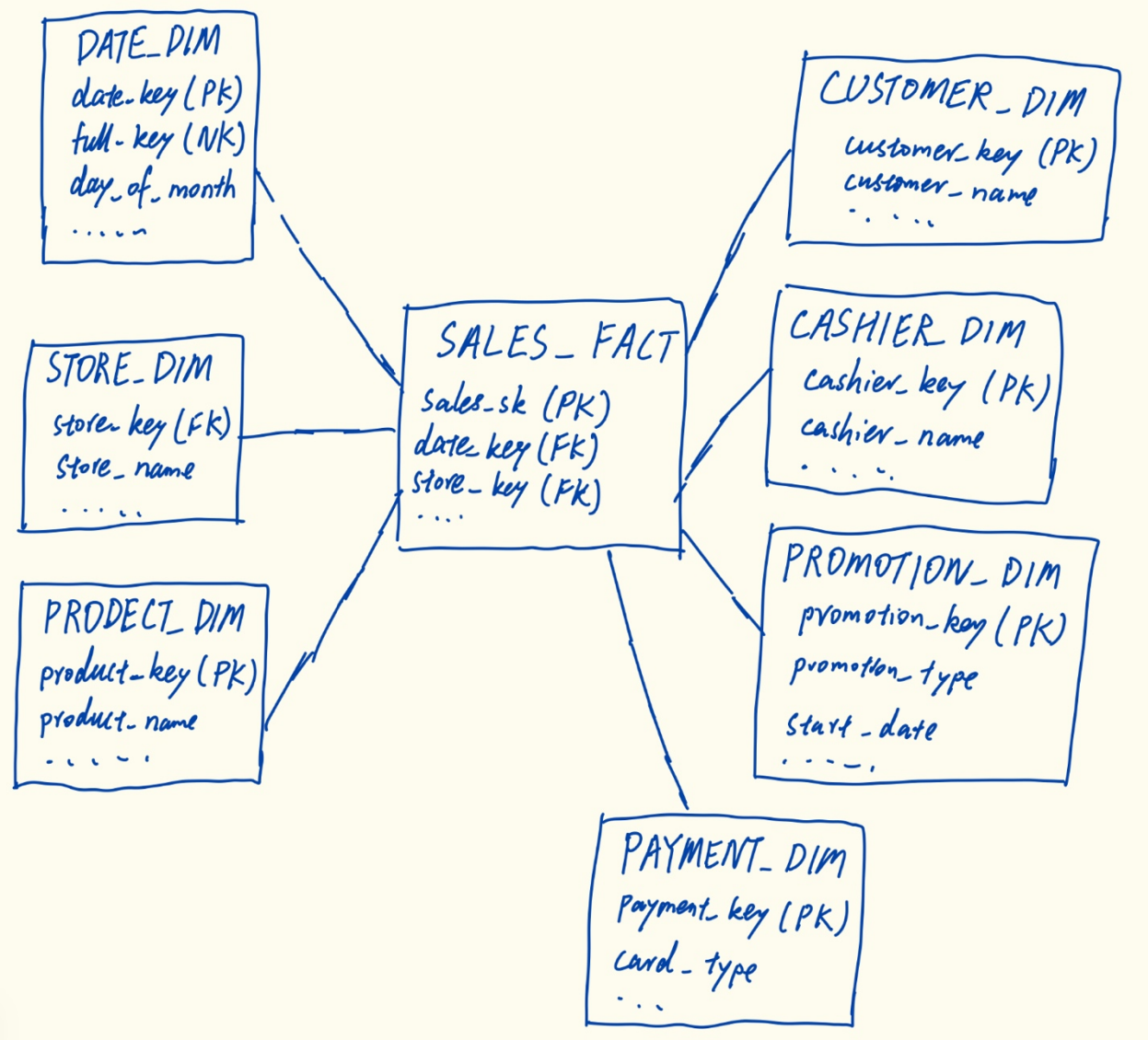
1. What customer information can be captured at the POS and how it can be handled for transactions?
2. How product attributes are stored in the POS which might not show on the receipt that enhance the sales analysis?
3. Could the POS support split payments of transactions, and if so, how could this be modeled?
4. How are promotions tracked in the POS at the transaction level, line-item level, or both and what attributes are available? Like promotion type and duration?

Dimensional Data Model

- Surrogate keys (PK): Integer-based primary keys in dimensions for performance and handling slowly changing dimensions (SCDs).
- Natural keys (NK): Business identifiers.
- Foreign keys (FK): In the fact table linking to dimension PKs (one-to-many relationships).
- Hierarchies: In dimensions (e.g., Date: year > quarter > month; Product: category > subcategory).
- Degenerate dimensions: Transaction_ID and Line_Number stored directly in the fact (low cardinality, no full dimension needed).

- xMeasures: Additive facts (e.g., quantity_sold) derived from receipt lines (e.g., extended_price = quantity_sold * unit_price).
- Attributes are based on typical receipt elements and extended with logical additions for analysis (justified in assumptions).

The diagram does not show all attribute names since it is too messy, instead, I put all of them in tables (Fact and Dimension) after the model to complement.



SALES_FACT (Fact Table)

Column Name	Type/Role	Description
sales_sk	PK	Unique surrogate key for each fact row.
date_key	FK	References DATE_DIM.
store_key	FK	References STORE_DIM.
product_key	FK	References PRODUCT_DIM.

Column Name	Type/Role	Description
customer_key	FK	References CUSTOMER_DIM.
cashier_key	FK	References CASHIER_DIM.
promotion_key	FK	References PROMOTION_DIM.
Payment_key	FK	References PAYMENT_DIM.
transaction_id	NK	Operational transaction number from the receipt.
line_item_number	NK	Sequence number for the line item on the receipt.
quantity_sold	Measure	Number of units sold.
unit_price	Measure	Price per unit from the receipt.
extended_price	Measure	Quantity * unit_price (line total before discount).
discount_amount	Measure	Any discount applied to the line item.
tax_amount	Measure	Tax for the line item.
net_amount	Measure	Extended price minus discount plus tax.

DATE_DIM (Dimension Table)

Column Name	Type/Role	Description
date_key	PK	Unique surrogate key.
full_date	NK	Actual date.
day_of_month	Attribute	Day number.
day_of_week	Attribute	Weekday name.
month_name	Attribute	Month name.
quarter	Attribute	Quarter number.
year	Attribute	Year.
is_holiday	Attribute	Flag for holiday (Y/N).
fiscal_year	Attribute	Fiscal year (assumed calendar match).

STORE_DIM (Dimension Table)

Column Name	Type/Role	Description
store_key	PK	Unique surrogate key.
store_number	NK	Store ID from POS system.
store_name	Attribute	Name.
address	Attribute	Street address.
city	Attribute	City.
state	Attribute	State.
zip_code	Attribute	ZIP code.
phone_number	Attribute	Store phone.

Column Name	Type/Role	Description
manager_name	Attribute	Store manager.
open_date	Attribute	Date store opened.
region	Attribute	Geographic region (assumed hierarchy).

PRODUCT_DIM (Dimension Table)

Column Name	Type/Role	Description
product_key	PK	Unique surrogate key.
sku	NK	Stock keeping unit/code from receipt.
product_name	Attribute	Name.
brand_name	Attribute	Brand.
category	Attribute	Product category.
sub_category	Attribute	Subcategory.
package_size	Attribute	Size.
unit_cost	Attribute	Cost to store (for margin analysis).
shelf_life_days	Attribute	Days until expiration.
is_perishable	Attribute	Flag for perishable items (Y/N).

CUSTOMER_DIM (Dimension Table)

Column Name	Type/Role	Description
customer_key	PK	Unique surrogate key.
customer_id	NK	Loyalty card or customer number.
customer_name	Attribute	Full name.
address	Attribute	Home address.
city	Attribute	City.
state	Attribute	State.
zip_code	Attribute	ZIP code.
phone_number	Attribute	Phone.
email	Attribute	Email address.
join_date	Attribute	Date joined loyalty program.
customer_segment	Attribute	Segment (e.g., Frequent Shopper; assumed).

CASHIER_DIM (Dimension Table)

Column Name	Type/Role	Description
cashier_key	PK	Unique surrogate key.
cashier_id	NK	Employee ID from POS.
cashier_name	Attribute	Full name.

Column Name	Type/Role	Description
hire_date	Attribute	Date hired.
position	Attribute	Role (e.g., Cashier, Supervisor).
department	Attribute	Department.

PROMOTION_DIM (Dimension Table)

Column Name	Type/Role	Description
promotion_key	PK	Unique surrogate key.
promotion_id	NK	Promotion code from POS.
promotion_name	Attribute	Name.
promotion_type	Attribute	Type.
discount_percent	Attribute	Percentage off.
start_date	Attribute	Promotion start date.
end_date	Attribute	Promotion end date.
media_type	Attribute	Advertising method.

PAYMENT_DIM (Dimension Table)

Column Name	Type/Role	Description
payment_key	PK	Unique surrogate key.
payment_method_id	NK	Code from POS.
payment_method_name	Attribute	e.g., Cash, Credit Card, Debit.
card_type	Attribute	e.g., Visa, Mastercard (if applicable).